LEARNING HUB MASTER SERIES

Linear Algebra: Orthogonality, Least Squares & QR Decomposition

Geometric Foundations, The Four Subspaces, Rigorous Least Squares Derivations, The A = QR Factorization, Spectral Theorem, SVD, and Hand Calculation Fast-Tracks

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Synthesized From:

• Gilbert Strang — *Introduction to Linear Algebra* (5th Edition)

• Kenji Hiranabe — *The Art of Linear Algebra* (Visual Matrix Patterns & Factorizations)

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Includes: 10 Hand-Calculation Fast-Track Shortcuts • 5 Step-by-Step Worked Problems

Table of Contents

Part I: Geometric Foundations & The Four Subspaces

Chapter 1: Dot Products, Norms, Angles & Cauchy-Schwarz — TRICKS 1 & 2

Chapter 2: The Four Fundamental Subspaces & Orthogonal Complements

Part II: Orthogonal Projections & Least Squares Deep-Dive

Chapter 3: Orthogonal Projections onto Lines & Subspaces — TRICKS 3 & 8

Chapter 4: Least Squares Approximations, Normal Equations & Regression — TRICKS 4 & 5

Part III: The A = QR Decomposition & Orthonormal Systems

Chapter 5: Orthonormal Bases & Square Orthogonal Matrices Q

Chapter 6: The Gram-Schmidt Process & A = QR Decomposition — TRICK 6

Chapter 7: Solving Least Squares via QR & Numerical Stability Analysis — TRICK 7

Part IV: Connected Factorizations (Spectral Theorem & SVD)

Chapter 8: Real Symmetric Matrices & The Spectral Theorem $S = Q\Lambda Q^T$ — TRICKS 9 & 10

Chapter 9: The Singular Value Decomposition $A = U\Sigma V^T$ & Pseudoinverse A^+

Part V: Step-by-Step Worked Problems & Reference Appendices

Chapter 10: 5 Complete Worked Problems with Hand-Calculation Fast-Tracks

Appendix A: Hiranabe's Five Factorizations & The Matrix World Hierarchy

Appendix B: Master Hand-Calculation Cheat Sheet & Formula Summary

Part I: Geometric Foundations & The Four Subspaces

Chapter 1: Dot Products, Norms, Angles & Cauchy-Schwarz

The inner product (dot product) is the geometric engine of linear algebra. It transforms raw algebraic vectors into geometric quantities: lengths, distances, angles, and above all, **perpendicularity (orthogonality)**.

1.1 Inner Products and Euclidean Length

For vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$, the dot product is defined algebraically as the sum of coordinate products, written equivalently in matrix form as a row vector multiplied by a column vector:

$$\mathbf{u} \cdot \mathbf{v} = \mathbf{u}^T \mathbf{v} = u_1v_1 + u_2v_2 + \ldots + u_nv_n = \sum_{i=1}^{n} u_iv_i$$

The **Euclidean length (norm)** $\|\mathbf{v}\|$ is the square root of the dot product of a vector with itself:

$$\|\mathbf{v}\| = \sqrt{(\mathbf{v}^T \mathbf{v})} = \sqrt{(v_1^2 + v_2^2 + \ldots + v_n^2)}$$

A vector is a **unit vector** if $\|\mathbf{u}\| = 1$. Any non-zero vector $\mathbf{v}$ is normalized into a unit direction vector by dividing by its length: $\mathbf{u} = \mathbf{v} / \|\mathbf{v}\|$.

▶ Orthogonality & The n-Dimensional Pythagorean Theorem

Two vectors $\mathbf{u}$ and $\mathbf{v}$ are **orthogonal** (perpendicular, written $\mathbf{u} \perp \mathbf{v}$) if and only if their dot product vanishes:

$$\mathbf{u} \perp \mathbf{v} \iff \mathbf{u}^T \mathbf{v} = 0$$

The Pythagorean Theorem: Vectors $\mathbf{u}$ and $\mathbf{v}$ satisfy $\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$ if and only if $\mathbf{u}^T \mathbf{v} = 0$.

◆ **Proof of the n-Dimensional Pythagorean Theorem**

Expand the squared length using matrix transpose rules:

$$\|\mathbf{u}+\mathbf{v}\|^2 = (\mathbf{u}+\mathbf{v})^T(\mathbf{u}+\mathbf{v}) = \mathbf{u}^T\mathbf{u} + \mathbf{u}^T\mathbf{v} + \mathbf{v}^T\mathbf{u} + \mathbf{v}^T\mathbf{v}$$

Since $\mathbf{u}^T\mathbf{v}$ is a scalar, $\mathbf{v}^T\mathbf{u} = \mathbf{u}^T\mathbf{v}$, yielding:

$$\|\mathbf{u}+\mathbf{v}\|^2 = \|\mathbf{u}\|^2 + 2(\mathbf{u}^T\mathbf{v}) + \|\mathbf{v}\|^2$$

Therefore, $\|\mathbf{u}+\mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$ precisely when the cross-term $2(\mathbf{u}^T\mathbf{v}) = 0$, proving that orthogonality is strictly equivalent to the Pythagorean relation in any dimension n.

1.2 Angles, Cauchy-Schwarz, and the Triangle Inequality

For any non-zero vectors in $\mathbb{R}^n$, the angle θ between them satisfies:

$$\cos\theta = \frac{\mathbf{u}^T\mathbf{v}}{\|\mathbf{u}\|\,\|\mathbf{v}\|}, \quad \text{where } 0 \le \theta \le \pi$$

▶ **The Cauchy-Schwarz & Triangle Inequalities**

For all vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$:

1. **Cauchy-Schwarz Inequality:** $|\mathbf{u}^T\mathbf{v}| \le \|\mathbf{u}\|\,\|\mathbf{v}\|$ (Equality holds iff **u** and **v** are collinear).
2. **Triangle Inequality:** $\|\mathbf{u}+\mathbf{v}\| \le \|\mathbf{u}\| + \|\mathbf{v}\|$.

TRICK 1 **Lightning Dot Products & Zero-Sum Cancellation** **2x Faster**

Standard Method (Slow): Multiplying coordinates blindly in linear order: $u_1v_1 + u_2v_2 + u_3v_3 + \ldots$, leading to large intermediate sums and mental fatigue.

⚡ **Fast Hand Calculation Shortcut: Visual Zero-Sum Pairing:** Scan coordinates first for 0, ±1, and opposite signs! Group terms that cancel to zero immediately. For example, to compute [3, −4, 5, 2] • [2, 3, −1, 7]:
• Pair 1: 3(2) = +6
• Pair 2: (−4)(3) = −12
• Pair 3: 5(−1) = −5
• Pair 4: 2(7) = +14
Notice (+6 − 5) = +1, and (+14 − 12) = +2 → Result is **+3** in 3 seconds!

TRICK 2 **Integer Vector Norm Scaling via GCD Factoring** **3x Faster**

Standard Method (Slow): Squaring large vector coordinates directly: $\|[12, 16, 20]^T\| = \sqrt{(144 + 256 + 400)} = \sqrt{800} = 20\sqrt{2}$. Highly error-prone under exam conditions.

⚡ **Fast Hand Calculation Shortcut: Factor out the Greatest Common Divisor (GCD) BEFORE squaring:**
$[12, 16, 20]^T = 4 \times [3, 4, 5]^T$.
Calculate the reduced norm: $\|[3, 4, 5]\| = \sqrt{(9 + 16 + 25)} = \sqrt{50} = 5\sqrt{2}$.
Multiply back the scalar: $4 \times 5\sqrt{2} = \mathbf{20\sqrt{2}}$. Eliminates squaring 3-digit numbers!

Chapter 2: The Four Fundamental Subspaces & Orthogonal Complements

Every m × n matrix A induces four fundamental subspaces that form the absolute cornerstone of linear algebra (Strang Sec 4.1; Hiranabe Figures 5 & 20). Orthogonality establishes the perfect geometric duality between these subspaces.

2.1 Definition of the Four Fundamental Subspaces

Subspace	Notation	Ambience	Dimension	Geometric Definition
Row Space	$C(A^T)$	$\mathbb{R}^n$	r	All linear combinations of the rows of A
Nullspace (Kernel)	$N(A)$	$\mathbb{R}^n$	$n - r$	All solutions to $A\mathbf{x} = \mathbf{0}$
Column Space (Range)	$C(A)$	$\mathbb{R}^m$	r	All linear combinations of the columns of A
Left Nullspace	$N(A^T)$	$\mathbb{R}^m$	$m - r$	All solutions to $A^T\mathbf{y} = \mathbf{0}$ (or $\mathbf{y}^TA = \mathbf{0}^T$)

▶ The Fundamental Theorem of Linear Algebra (Parts 1 & 2)

For any m × n matrix A of rank r:

1. **In $\mathbb{R}^n$:** The nullspace $N(A)$ is the **orthogonal complement** of the row space $C(A^T)$:

$$N(A) = (C(A^T))^{\perp}, \quad \dim C(A^T) + \dim N(A) = r + (n - r) = n$$

2. **In $\mathbb{R}^m$:** The left nullspace $N(A^T)$ is the **orthogonal complement** of the column space $C(A)$:

$$N(A^T) = (C(A))^{\perp}, \quad \dim C(A) + \dim N(A^T) = r + (m - r) = m$$

◆ Why Row Space is Perpendicular to Nullspace (C(AT) ⊥ N(A))

Let $\mathbf{x} \in N(A)$, so by definition $A\mathbf{x} = \mathbf{0}$. Written row-by-row:

$$\begin{vmatrix} \text{row 1 of A} & \cdot & \mathbf{x} \\ \text{row 2 of A} & \cdot & \mathbf{x} \\ \vdots & \vdots & \vdots \\ \text{row m of A} & \cdot & \mathbf{x} \end{vmatrix} = \begin{vmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{vmatrix}$$

Every single row $\mathbf{r}_i$ of A has dot product zero with $\mathbf{x}$: $\mathbf{r}_i^T\mathbf{x} = 0$.

Any vector $\mathbf{v}$ in the row space $C(A^T)$ is a linear combination of rows: $\mathbf{v} = \sum c_i\mathbf{r}_i$.

Therefore, $\mathbf{v}^T\mathbf{x} = (\sum c_i\mathbf{r}_i)^T\mathbf{x} = \sum c_i(\mathbf{r}_i^T\mathbf{x}) = \sum c_i(0) = 0$.

Since every row space vector is perpendicular to every nullspace vector, $C(A^T) \perp N(A)$.

◎ Strang's Big Picture & Hiranabe's Mapping

Every vector $\mathbf{x} \in \mathbb{R}^n$ decomposes uniquely into orthogonal components:

$$\mathbf{x} = \mathbf{x}_{row} + \mathbf{x}_{null}, \quad \text{where } \mathbf{x}_{row} \in C(A^T), \ \ \mathbf{x}_{null} \in N(A)$$

Multiplying by A yields: $A\mathbf{x} = A(\mathbf{x}_{row} + \mathbf{x}_{null}) = A\mathbf{x}_{row} + \mathbf{0} = A\mathbf{x}_{row} \in C(A)$.

Fundamental Insight: The nullspace is crushed entirely to **0**. The matrix A acts as a **one-to-one, invertible linear bijection** between the row space $C(A^T)$ and the column space C(A)!

Part II: Orthogonal Projections & Least Squares Deep-Dive

Chapter 3: Orthogonal Projections onto Lines & Subspaces

When a linear system $A\mathbf{x} = \mathbf{b}$ has no solution because $\mathbf{b}$ lies outside the column space C(A), the best approximation is obtained by **projecting b orthogonally onto C(A)**.

3.1 1D Projection onto a Line

Let $\mathbf{a} \neq \mathbf{0}$ be a direction vector in $\mathbb{R}^m$. We seek the point $\mathbf{p} = \hat{x}\mathbf{a}$ on the line spanned by $\mathbf{a}$ that is closest to $\mathbf{b}$. The key geometric requirement is that the error vector $\mathbf{e} = \mathbf{b} - \mathbf{p}$ must be perpendicular to $\mathbf{a}$:

$$\mathbf{a} \perp (\mathbf{b} - \hat{x}\mathbf{a}) \;\Rightarrow\; \mathbf{a}^T(\mathbf{b} - \hat{x}\mathbf{a}) = 0 \;\Rightarrow\; \mathbf{a}^T\mathbf{b} - \hat{x}(\mathbf{a}^T\mathbf{a}) = 0$$

Solving for the scalar coefficient $\hat{x}$ and the projection vector $\mathbf{p}$:

$$\hat{x} = \frac{\mathbf{a}^T\mathbf{b}}{\mathbf{a}^T\mathbf{a}}, \qquad \mathbf{p} = \hat{x}\mathbf{a} = \frac{\mathbf{a}^T\mathbf{b}}{\mathbf{a}^T\mathbf{a}}\,\mathbf{a} = \frac{\mathbf{a}\mathbf{a}^T}{\mathbf{a}^T\mathbf{a}}\,\mathbf{b}$$

The rank-1 **projection matrix** onto the line is $P = \frac{\mathbf{a}\mathbf{a}^T}{\mathbf{a}^T\mathbf{a}}$.

3.2 Subspace Projection onto C(A)

Now let $A = [\mathbf{a}_1 \dots \mathbf{a}_n] \in \mathbb{R}^{m \times n}$ with linearly independent columns (rank $r = n \leq m$). The projection $\mathbf{p} \in C(A)$ is a linear combination of the columns: $\mathbf{p} = A\hat{x} = \hat{x}_1\mathbf{a}_1 + \dots + \hat{x}_n\mathbf{a}_n$. The error vector $\mathbf{e} = \mathbf{b} - A\hat{x}$ must be perpendicular to the *entire* column space C(A):

$$\mathbf{e} \perp C(A) \;\Leftrightarrow\; \mathbf{e} \in N(A^T) \;\Leftrightarrow\; A^T(\mathbf{b} - A\hat{x}) = \mathbf{0}$$

Distributing A^T yields the historic **Normal Equations**:

$$A^TA\hat{x} = A^T\mathbf{b}$$

▶ Invertibility of ATA & The Subspace Projection Matrix

Theorem: The square matrix $A^TA \in \mathbb{R}^{n \times n}$ is invertible if and only if A has linearly independent columns (nullspace N(A) = {**0**}).
When A has independent columns, the optimal coordinate vector $\hat{x}$ and projection matrix P are:

$$\hat{x} = (A^TA)^{-1} A^T\mathbf{b}, \quad \mathbf{p} = A\hat{x} = A(A^TA)^{-1} A^T\mathbf{b} = P\mathbf{b}$$

$$P = A(A^TA)^{-1} A^T$$

◆ Proof that N(ATA) = N(A) (Invertibility of ATA)

We must show that $A^TA\mathbf{x} = \mathbf{0}$ implies $\mathbf{x} = \mathbf{0}$.

1. If $A\mathbf{x} = \mathbf{0}$, then multiplying by A^T gives $A^T(A\mathbf{x}) = A^T\mathbf{0} = \mathbf{0}$, so $N(A) \subseteq N(A^TA)$.
2. Conversely, suppose $A^TA\mathbf{x} = \mathbf{0}$. Multiply on the left by $\mathbf{x}^T$:

$$\mathbf{x}^T(A^TA\mathbf{x}) = \mathbf{x}^T\mathbf{0} = 0 \;\Rightarrow\; (A\mathbf{x})^T(A\mathbf{x}) = 0 \;\Rightarrow\; \|A\mathbf{x}\|^2 = 0$$

3. Since the norm of a vector is zero only if the vector itself is zero, $A\mathbf{x} = \mathbf{0}$.
4. Since A has independent columns, $A\mathbf{x} = \mathbf{0}$ requires $\mathbf{x} = \mathbf{0}$.

Thus $N(A^TA) = \{\mathbf{0}\}$, proving that the n × n matrix A^TA is strictly non-singular and invertible!

3.3 Properties of Projection Matrices

- **Symmetry:** $P^T = (A(A^TA)^{-1}A^T)^T = (A^T)^T ((A^TA)^{-1})^T A^T = A(A^TA)^{-1}A^T = P$.
- **Idempotence:** $P^2 = A(A^TA)^{-1}(A^TA)(A^TA)^{-1}A^T = A(A^TA)^{-1}A^T = P$. (Projecting twice does nothing new).
- **Complementary Projection:** $(I - P)$ is also symmetric and idempotent. It projects **b** onto the orthogonal complement $N(A^T)$: $\mathbf{e} = (I - P)\mathbf{b}$. Every vector decomposes cleanly into $\mathbf{b} = P\mathbf{b} + (I - P)\mathbf{b} = \mathbf{p} + \mathbf{e}$.

TRICK 3 The Swap-Negate-Divide Rule for 2 × 2 Matrix Inversion

5 Seconds

Standard Method (Slow): Setting up a 2 × 4 augmented matrix [M | I] and performing Gauss-Jordan row elimination with row multiplications and fractions.

⚡ **Fast Hand Calculation Shortcut: Memorize the Swap-Negate-Divide Mnemonic:**

For any 2 × 2 matrix $M = \begin{vmatrix} a & b \\ c & d \end{vmatrix}$:

1. Compute determinant: $\Delta = ad - bc$.
2. **Swap** the main diagonal entries: $a \leftrightarrow d$.
3. **Negate** the off-diagonal entries: $b \to -b$, $c \to -c$.
4. **Divide** everything by Δ:

$$M^{-1} = \frac{1}{ad - bc} \begin{vmatrix} d & -b \\ -c & a \end{vmatrix}$$

Takes exactly 5 seconds in your head!

TRICK 8 Decoupled Subspace Projection with Mutually Orthogonal Columns

Instant

Standard Method (Slow): Forming the 2 × 2 matrix A^TA, computing its inverse $(A^TA)^{-1}$, and multiplying out $P = A(A^TA)^{-1}A^T$.

⚡ **Fast Hand Calculation Shortcut: If columns a_1 and a_2 are orthogonal ($a_1^Ta_2 = 0$):**

The off-diagonal entries of A^TA vanish completely! A^TA is strictly diagonal:

$$A^TA = \begin{vmatrix} a_1^Ta_1 & 0 \\ 0 & a_2^Ta_2 \end{vmatrix}$$

The projection splits into two independent 1D line projections with ZERO matrix inversion needed:

$$\mathbf{p} = \frac{a_1^T\mathbf{b}}{a_1^Ta_1}\mathbf{a}_1 + \frac{a_2^T\mathbf{b}}{a_2^Ta_2}\mathbf{a}_2$$

Saves 80% of computation time!

Chapter 4: Least Squares Approximations, Normal Equations & Regression

In physical experiments and data science, systems $A\mathbf{x} = \mathbf{b}$ have far more equations than unknowns ($m > n$). Measurement noise ensures that $\mathbf{b} \notin C(A)$. No exact solution exists. The **method of least squares** finds the candidate $\hat{x}$ that minimizes the total squared Euclidean error:

$$E(\mathbf{x}) = \|A\mathbf{x} - \mathbf{b}\|^2 = \sum_{i=1}^{m} (a_{i1}x_1 + \ldots + a_{in}x_n - b_i)^2$$

4.1 Dual Independent Derivations of the Normal Equations

▶ Dual Derivations of the Normal Equations: $A^TA\,\hat{x} = A^Tb$

The optimal least squares solution $\hat{x}$ satisfies the fundamental system:

$$A^TA\,\hat{x} = A^T\mathbf{b}$$

Below are two completely independent derivations: one via **Geometric Projection**, and one via **Multivariable Calculus**.

◆ Derivation 1: Geometric Projection (Pythagorean Theorem)

Let $\mathbf{p} = A\,\hat{x}$ be the orthogonal projection of $\mathbf{b}$ onto $C(A)$, so $\mathbf{e} = \mathbf{b} - \mathbf{p} \perp C(A)$.

For *any* candidate vector $\mathbf{x} \in \mathbb{R}^n$, the error can be decomposed as:

$$A\mathbf{x} - \mathbf{b} = (A\mathbf{x} - \mathbf{p}) + (\mathbf{p} - \mathbf{b}) = (A\mathbf{x} - \mathbf{p}) - \mathbf{e}$$

Notice that $(A\mathbf{x} - \mathbf{p})$ lies entirely inside $C(A)$, while $\mathbf{e}$ is perpendicular to $C(A)$.
By the Pythagorean Theorem:

$$\|A\mathbf{x} - \mathbf{b}\|^2 = \|A\mathbf{x} - \mathbf{p}\|^2 + \|\mathbf{e}\|^2$$

Since $\|\mathbf{e}\|^2$ is fixed and independent of $\mathbf{x}$, the sum is strictly minimized when $\|A\mathbf{x} - \mathbf{p}\|^2 = 0$, which occurs if and only if $A\mathbf{x} = \mathbf{p}$.

Since $\mathbf{e} = \mathbf{b} - A\,\hat{x} \perp C(A)$, it lies in $N(A^T)$, which means:

$$A^T(\mathbf{b} - A\,\hat{x}) = \mathbf{0} \;\Rightarrow\; A^TA\,\hat{x} = A^T\mathbf{b}$$

◆ Derivation 2: Multivariable Calculus (Gradient Optimization)

Expand the quadratic error function $E(\mathbf{x})$ using vector-matrix multiplication:

$$E(\mathbf{x}) = (A\mathbf{x} - \mathbf{b})^T(A\mathbf{x} - \mathbf{b}) = \mathbf{x}^TA^TA\mathbf{x} - 2\mathbf{x}^TA^T\mathbf{b} + \mathbf{b}^T\mathbf{b}$$

To find the minimum, compute the gradient vector $\nabla_{\mathbf{x}} E$ with respect to $\mathbf{x}$:

$$\nabla_{\mathbf{x}}(\mathbf{x}^TA^TA\mathbf{x}) = 2A^TA\mathbf{x}, \quad \nabla_{\mathbf{x}}(2\mathbf{x}^TA^T\mathbf{b}) = 2A^T\mathbf{b}$$

Setting the gradient to the zero vector $\nabla_{\mathbf{x}} E = \mathbf{0}$:

$$2A^TA\mathbf{x} - 2A^T\mathbf{b} = \mathbf{0} \;\Rightarrow\; A^TA\,\hat{x} = A^T\mathbf{b}$$

The Hessian matrix $\nabla^2 E = 2A^TA$ is strictly positive definite (when A has independent columns), guaranteeing that $\hat{x}$ is a global minimum!

4.2 Fitting a Straight Line (y = C + Dt)

Given m data points $(t_1, b_1), (t_2, b_2), \ldots, (t_m, b_m)$, we seek the line $b = C + Dt$ that minimizes the sum of squared vertical errors $\sum (C + Dt_i - b_i)^2$. In matrix form $A\mathbf{x} = \mathbf{b}$:

$$\begin{vmatrix} 1 & t_1 \\ 1 & t_2 \\ \vdots & \vdots \\ 1 & t_m \end{vmatrix} \begin{vmatrix} C \\ D \end{vmatrix} = \begin{vmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{vmatrix}$$

The Normal Equations system $A^TA\,\hat{x} = A^T\mathbf{b}$ expands to:

$$\begin{vmatrix} m & \sum t_i \\ \sum t_i & \sum t_i^2 \end{vmatrix} \begin{vmatrix} C \\ D \end{vmatrix} = \begin{vmatrix} \sum b_i \\ \sum t_i b_i \end{vmatrix}$$

TRICK 4 Symmetric Reflection Shortcut for Forming A^TA

50% Faster

Standard Method (Slow): Computing all n × n dot products individually. For a 3 × 3 matrix, this requires 9 full vector multiplications.

⚡ **Fast Hand Calculation Shortcut: A^TA is ALWAYS symmetric: $(A^TA)^T = A^TA$.**
Never compute below the diagonal! Compute only:

1. The n diagonal dot products (column lengths squared $\|\mathbf{a}_i\|^2$).
2. The $n(n-1)/2$ strictly upper-triangular entries ($\mathbf{a}_i^T\mathbf{a}_j$ for $i < j$).
3. Simply **copy/reflect** the upper entries into the lower positions.

Saves exactly 50% of off-diagonal multiplications!

TRICK 5 The Centered Data Secret Weapon (Decoupling Slope & Intercept)

75% Faster

Standard Method (Slow): Solving the full coupled 2 × 2 normal equation matrix with determinants, fractions, and large sums of t_i.

⚡ **Fast Hand Calculation Shortcut: Shift the time predictor to its mean: $t'_i = t_i - \bar{t}$**, where $\bar{t} = (1/m)\sum t_i$.
By definition of the mean, **$\sum t'_i = 0$!**

The off-diagonal entries in A^TA become zero. A^TA diagonalizes completely:

$$\begin{bmatrix} m & 0 \\ 0 & \sum (t'_i)^2 \end{bmatrix} \begin{bmatrix} C' \\ D \end{bmatrix} = \begin{bmatrix} \sum b_i \\ \sum t'_i b_i \end{bmatrix}$$

The intercept and slope decouple instantly with ZERO matrix inversion:

$$C' = \bar{b} = \frac{1}{m}\sum b_i, \qquad D = \frac{\sum t'_i\, b_i}{\sum (t'_i)^2}$$

Original intercept is simply recovered as: **$C = C' - D\,\bar{t}$**. Cuts calculation time by 75%!

Part III: The A = QR Decomposition & Orthonormal Systems

Chapter 5: Orthonormal Bases & Square Orthogonal Matrices Q

The ideal basis in linear algebra consists of vectors that are mutually perpendicular and normalized to length 1.

5.1 Orthonormal Vectors & The Matrix Q

Vectors $\mathbf{q}_1, \mathbf{q}_2, \ldots, \mathbf{q}_n \in \mathbb{R}^m$ are **orthonormal** if:

$$\mathbf{q}_i^T \mathbf{q}_j = \delta_{ij} = \{ 1 \text{ if } i = j, \quad 0 \text{ if } i \neq j \}$$

Placing these vectors into the columns of an m × n matrix $Q = [\mathbf{q}_1 \ldots \mathbf{q}_n]$:

$$Q^T Q = \begin{bmatrix} \mathbf{q}_1^T \\ \vdots \\ \mathbf{q}_n^T \end{bmatrix} [\mathbf{q}_1 \ldots \mathbf{q}_n] = \begin{bmatrix} 1 & 0 & \ldots & 0 \\ 0 & 1 & \ldots & 0 \\ \vdots & \vdots & \&ddots; & \vdots \\ 0 & 0 & \ldots & 1 \end{bmatrix} = I_n$$

▶ Square Orthogonal Matrices (m = n)

When Q is square (n × n), having orthonormal columns makes Q an **orthogonal matrix**:

1. **Instant Inverse:** $Q^T Q = I \Rightarrow \mathbf{Q^{-1} = Q^T}$ and $\mathbf{QQ^T = I}$.
2. **Isometry (Length Preservation):** $\|Q\mathbf{x}\| = \|\mathbf{x}\|$ for all $\mathbf{x}$.
3. **Angle Preservation:** $(Q\mathbf{x})^T(Q\mathbf{y}) = \mathbf{x}^T Q^T Q \mathbf{y} = \mathbf{x}^T \mathbf{y}$.
4. **Determinant:** $\det(Q) = \pm 1$ (Rotations have det = +1; Reflections have det = −1).

5.2 Decoupled Subspace Projections with Orthonormal Columns

When a subspace has an orthonormal basis Q, the general projection formula $P = A(A^TA)^{-1}A^T$ collapses into sheer elegance:

$$P = Q(Q^TQ)^{-1}Q^T = Q(I)^{-1}Q^T = \mathbf{Q\,Q^T}$$

Written as a sum of rank-1 outer product projection matrices (Hiranabe Pattern P4):

$$P = [\mathbf{q}_1 \dots \mathbf{q}_n] \{mat([['\mathbf{q}_1{}^T'], [' \vdots '], ['\mathbf{q}_n{}^T']])\} = \mathbf{q}_1\mathbf{q}_1{}^T + \mathbf{q}_2\mathbf{q}_2{}^T + \dots + \mathbf{q}_n\mathbf{q}_n{}^T$$

The projection of any vector $\mathbf{b}$ is simply the sum of its 1D projections onto each coordinate axis independently:

$$\mathbf{p} = P\mathbf{b} = (\mathbf{q}_1{}^T\mathbf{b})\mathbf{q}_1 + (\mathbf{q}_2{}^T\mathbf{b})\mathbf{q}_2 + \dots + (\mathbf{q}_n{}^T\mathbf{b})\mathbf{q}_n$$

Chapter 6: The Gram-Schmidt Process & A = QR Decomposition

The Gram-Schmidt algorithm takes any set of linearly independent column vectors $\mathbf{a}_1, \dots, \mathbf{a}_n$ and systematically strips away redundant directional components to produce an orthonormal basis $\mathbf{q}_1, \dots, \mathbf{q}_n$.

6.1 The Classical Gram-Schmidt Algorithm

1. **Step 1:** Start with $\mathbf{a}_1$. Orthogonal direction: $\mathbf{v}_1 = \mathbf{a}_1$. Unit vector: $\mathbf{q}_1 = \mathbf{v}_1 / \|\mathbf{v}_1\|$.
2. **Step 2:** Subtract the projection of $\mathbf{a}_2$ along $\mathbf{q}_1$:

$$\mathbf{v}_2 = \mathbf{a}_2 - (\mathbf{q}_1{}^T\mathbf{a}_2)\mathbf{q}_1, \qquad \mathbf{q}_2 = \mathbf{v}_2 / \|\mathbf{v}_2\|$$

3. **Step k:** Subtract projections along all previously computed orthonormal vectors $\mathbf{q}_1, \dots, \mathbf{q}_{k-1}$:

$$\mathbf{v}_k = \mathbf{a}_k - \sum_{j=1}^{k-1} (\mathbf{q}_j{}^T\mathbf{a}_k)\mathbf{q}_j, \qquad \mathbf{q}_k = \mathbf{v}_k / \|\mathbf{v}_k\|$$

6.2 Matrix Factorization: A = QR

Rewriting the Gram-Schmidt relations expressing the original columns $\mathbf{a}_k$ in terms of $\mathbf{q}_j$:

$$\mathbf{a}_1 = (\mathbf{q}_1{}^T\mathbf{a}_1)\mathbf{q}_1 = r_{11}\mathbf{q}_1$$
$$\mathbf{a}_2 = (\mathbf{q}_1{}^T\mathbf{a}_2)\mathbf{q}_1 + (\mathbf{q}_2{}^T\mathbf{a}_2)\mathbf{q}_2 = r_{12}\mathbf{q}_1 + r_{22}\mathbf{q}_2$$
$$\mathbf{a}_k = r_{1k}\mathbf{q}_1 + r_{2k}\mathbf{q}_2 + \dots + r_{kk}\mathbf{q}_k$$

where $r_{jk} = \mathbf{q}_j{}^T\mathbf{a}_k$ for $j < k$, and $r_{kk} = \|\mathbf{v}_k\| > 0$. Collecting into matrix form (Hiranabe Pattern P1 & Figure 15):

$$[\mathbf{a}_1 \dots \mathbf{a}_n] = [\mathbf{q}_1 \dots \mathbf{q}_n] \begin{vmatrix} r_{11} & r_{12} & \dots & r_{1n} \\ 0 & r_{22} & \dots & r_{2n} \\ \vdots & \vdots & \&ddots; & \vdots \\ 0 & 0 & \dots & r_{nn} \end{vmatrix} \Rightarrow \mathbf{A = QR}$$

Notice that Q has orthonormal columns ($Q^TQ = I$) and R is **strictly upper-triangular** with positive diagonal entries $r_{ii} > 0$! Multiplying both sides by Q^T yields: $\mathbf{R = Q^TA}$.

TRICK 6 **Radical-Free Gram-Schmidt (The Integer Method)** **100% Radical-Free**

Standard Method (Slow): Normalizing at every single intermediate step: dividing by √3 or √14, causing ugly nested radicals in all subsequent dot products.

⚡ **Fast Hand Calculation Shortcut: Keep intermediate orthogonal vectors $\mathbf{v_k}$ as un-normalized integers!**
Project using the un-normalized formula:

$$\mathbf{v}_k = \mathbf{a}_k - \frac{\mathbf{v}_1^T\mathbf{a}_k}{\mathbf{v}_1^T\mathbf{v}_1}\mathbf{v}_1 - \dots - \frac{\mathbf{v}_{k-1}^T\mathbf{a}_k}{\mathbf{v}_{k-1}^T\mathbf{v}_{k-1}}\mathbf{v}_{k-1}$$

All dot products remain clean rational numbers! Clear any common denominator by scaling $\mathbf{v_k}$ by an integer (direction is unchanged).
Normalize ONLY ONCE at the very final step to get $\mathbf{q}_1, \dots, \mathbf{q}_n$! Completely eliminates radical arithmetic during the algorithm.

Chapter 7: Solving Least Squares via QR & Numerical Stability Analysis

The A = QR factorization represents the pinnacle of practical least squares computation. It bypasses the need to invert A^TA and provides bulletproof numerical stability.

7.1 The Great Normal Equations Simplification

Substitute A = QR into the Normal Equations $A^TA\,\hat{x} = A^T\mathbf{b}$:

$$(QR)^T(QR)\,\hat{x} = (QR)^T\mathbf{b} \;\Rightarrow\; (R^TQ^T)(QR)\,\hat{x} = R^TQ^T\mathbf{b}$$

Since $Q^TQ = I$, the middle terms cancel:

$$R^T(Q^TQ)R\,\hat{x} = R^TQ^T\mathbf{b} \;\Rightarrow\; R^TR\,\hat{x} = R^T(Q^T\mathbf{b})$$

Since A has linearly independent columns, the diagonal entries of R are strictly positive ($r_{ii} > 0$). Thus R is non-singular, and its transpose R^T is invertible. Multiplying on the left by $(R^T)^{-1}$:

▶ The QR Least Squares Master Formula

$$R\,\hat{x} = Q^T b$$

Key Takeaway: We NEVER form A^TA. We NEVER invert A^TA.
We simply compute the vector $\mathbf{d} = Q^T\mathbf{b}$, and solve $R\,\hat{x} = \mathbf{d}$ instantly via **back-substitution**!

TRICK 7 **Solving $R\,\hat{x} = Q^Tb$ via Fast Upper-Triangular Back-Substitution** **4x Faster**

Standard Method (Slow): Setting up augmented matrix $[R \mid Q^Tb]$ and performing full Gauss-Jordan elimination.

⚡ **Fast Hand Calculation Shortcut: Direct Back-Substitution:** Because R is upper-triangular, start at the bottom equation and climb upwards:
1. For 2 × 2 system {mat([['r_{11}', 'r_{12}'], [0, 'r_{22}']])} {col_vec('$\hat{x}_1$', '$\hat{x}_2$')} = {col_vec('d_1', 'd_2')}:
• $\hat{x}_2 = d_2 / r_{22}$
• $\hat{x}_1 = (d_1 - r_{12}\,\hat{x}_2) / r_{11}$
Takes 10 seconds of arithmetic!

7.2 Numerical Stability: Why QR Triumphs Over Normal Equations

In computational mathematics, the sensitivity of a linear system to round-off error is governed by its **condition number**:

$$\kappa(A) = \sigma_{max} / \sigma_{min} \geq 1$$

When solving least squares by forming the Normal Equations $A^TA\,\hat{x} = A^T\mathbf{b}$:

$$\kappa(A^TA) = (\kappa(A))^2$$

⚠ The Catastrophic Squaring of Condition Numbers

Squaring the condition number doubles the loss of significant digits:

Suppose a data matrix has condition number $\kappa(A) = 10^4$.
• Forming A^TA squares the condition number to $\kappa(A^TA) = 10^8$.
• In standard IEEE 754 single-precision floating point (approx. 7 decimal digits of precision), a condition number of 10^8 wipes out ALL significant digits! The computed solution $\hat{x}$ will be pure numerical garbage.

The QR Solution: Because Q is an isometry ($\kappa(Q) = 1$), $\kappa(R) = \kappa(A)$. Solving $R\,\hat{x} = Q^T\mathbf{b}$ preserves all original digits of accuracy!

Method	Condition Number	Flops ($m \gg n$)	Numerical Stability	Recommended Use Case
Normal Equations ($A^TA\,\hat{x} = A^Tb$)	$\kappa(A)^2$	$\sim mn^2 + n^3/3$	Poor (loss of precision on ill-conditioned data)	Well-conditioned, small hand calculations
Modified Gram-Schmidt (MGS)	$\kappa(A)$	$\sim 2mn^2$	Good (avoids squaring condition number)	Iterative solvers, teaching, general QR
Householder Reflections	$\kappa(A)$	$\sim 2mn^2 - 2n^3/3$	Excellent (strict machine-precision orthogonality)	Production scientific computing (LAPACK, NumPy)

Part IV: Connected Factorizations (Spectral Theorem & SVD)

Chapter 8: Real Symmetric Matrices & The Spectral Theorem $S = Q\Lambda Q^T$

Every symmetric matrix ($S = S^T$) possesses magical geometric properties dictated by orthogonality (Strang Sec. 6.4; Hiranabe Sec. 6.4).

▶ The Spectral Theorem for Real Symmetric Matrices

Every real symmetric matrix $S \in \mathbb{R}^{n \times n}$ has:

1. All n eigenvalues $\lambda_1, \ldots, \lambda_n$ are **strictly real numbers**.
2. Eigenvectors corresponding to distinct eigenvalues are **strictly orthogonal**.
3. There exists an orthonormal matrix $Q = [\mathbf{q}_1 \ \ldots \ \mathbf{q}_n]$ such that:

$$S = Q \Lambda Q^T, \quad \text{where } Q^{-1} = Q^T$$

8.1 Hiranabe's Pattern P4: Sum of Rank-1 Projections

Multiplying out $S = Q\Lambda Q^T$ using column-row multiplication (Hiranabe Pattern P4):

$$S = [\mathbf{q}_1 \ \ldots \ \mathbf{q}_n] \ \{\text{mat}([[\lambda_1, 0], [0, \lambda_n]])\} \ \{\text{mat}([[\mathbf{q}_1^T], [\mathbf{q}_n^T]])\} = \lambda_1 \mathbf{q}_1\mathbf{q}_1^T + \lambda_2 \mathbf{q}_2\mathbf{q}_2^T + \ldots + \lambda_n \mathbf{q}_n\mathbf{q}_n^T$$

Each term $P_i = \mathbf{q}_i\mathbf{q}_i^T$ is a rank-1 orthogonal projection matrix onto the eigenvector axis! **Any real symmetric matrix is nothing more than a weighted sum of mutually orthogonal projection matrices!**

TRICK 9 **2 × 2 Symmetric Eigenvalues via Trace-Determinant Mental Math** — Instant

Standard Method (Slow): Setting up $\det(S - \lambda I) = \lambda^2 - (a+d)\lambda + (ad - b^2) = 0$ and applying the full quadratic formula.

⚡ **Fast Hand Calculation Shortcut: Use the Average & Radius Shortcut:**
Let $m = \text{trace}(S)/2 = (a + d)/2$ (the midpoint).
Let $p = \det(S) = ad - b^2$.
Then: $\boldsymbol{\lambda = m \pm \sqrt{m^2 - p}}$.
For example, for $S = \{\text{mat}([[2, 1], [1, 2]])\}$: $m = 4/2 = 2$, $p = 4 - 1 = 3$.
$\lambda = 2 \pm \sqrt{(4 - 3)} = 2 \pm 1 \rightarrow \boldsymbol{\lambda_1 = 3, \lambda_2 = 1}$ in 3 seconds!

TRICK 10 **Instant 2 × 2 Symmetric Eigenvectors without Row Operations** **Zero Row Ops**

Standard Method (Slow): Row reducing $(S - \lambda I)$ to find nullspace vectors.

⚡ **Fast Hand Calculation Shortcut:** For symmetric S = {mat([['a', 'b'], ['b', 'd']])} and eigenvalue λ_1:

The first row of $(S - \lambda_1 I)$ is $[(a - \lambda_1), b]$.

A perpendicular vector is IMMEDIATELY: $\mathbf{x_1} = \begin{vmatrix} -b \\ a - \lambda_1 \end{vmatrix}$.

By the Spectral Theorem, the second eigenvector MUST be orthogonal, so rotate 90°: $\mathbf{x_2} = \begin{vmatrix} a - \lambda_1 \\ b \end{vmatrix}$.

Zero row operations needed!

Chapter 9: The Singular Value Decomposition $A = U\Sigma V^T$ & Pseudoinverse

While the Spectral Theorem diagonalizes square symmetric matrices using one orthonormal basis Q, the **Singular Value Decomposition (SVD)** diagonalizes ANY m × n matrix using TWO orthonormal bases (Strang Ch. 7; Hiranabe Sec. 6.5).

▶ The Fundamental Theorem of the SVD

Any real m × n matrix A of rank r factors into:

$$\mathbf{A = U \Sigma V^T}$$

where:

- $U \in \mathbb{R}^{m \times m}$ is orthogonal ($U^T U = I_m$): Orthonormal eigenvectors of AA^T.
- $V \in \mathbb{R}^{n \times n}$ is orthogonal ($V^T V = I_n$): Orthonormal eigenvectors of $A^T A$.
- $\Sigma \in \mathbb{R}^{m \times n}$ is diagonal with singular values: $\sigma_1 \geq \sigma_2 \geq \ldots \geq \sigma_r > 0$, where $\sigma_i = \sqrt{\lambda_i}(A^T A)$.

9.1 Connecting SVD to the Four Fundamental Subspaces

Basis Vectors	Matrix	Subspace Spanned	Dimension
$\mathbf{v}_1, \ldots, \mathbf{v}_r$	First r columns of V	**Row Space $C(A^T)$**	r
$\mathbf{v}_{r+1}, \ldots, \mathbf{v}_n$	Last n−r columns of V	**Nullspace N(A)**	n − r
$\mathbf{u}_1, \ldots, \mathbf{u}_r$	First r columns of U	**Column Space C(A)**	r
$\mathbf{u}_{r+1}, \ldots, \mathbf{u}_m$	Last m−r columns of U	**Left Nullspace $N(A^T)$**	m − r

The SVD diagonalizes matrix multiplication into r decoupled 1D equations: $\mathbf{A\, v_i = \sigma_i\, u_i}$ (for i = 1, ..., r).

9.2 The Moore-Penrose Pseudoinverse A^+

When A is not full column rank, A^TA is not invertible, and least squares has infinitely many solutions. The **pseudoinverse** finds the unique solution x^+ having **minimum Euclidean norm**:

$$A^+ = V\,\Sigma^+\,U^T = \sum_{i=1}^{r} \{frac('1', '\sigma_i')\}\,\mathbf{v}_i\mathbf{u}_i^T, \qquad \mathbf{x}^+ = A^+\mathbf{b} \in C(A^T)$$

Part V: Step-by-Step Worked Problems & Reference Appendices

Chapter 10: 5 Complete Worked Problems with Hand-Calculation Fast-Tracks

Problem 1: 3D Subspace Projection & Error Vector Check

Chapters 3 & Fast-Track 8

Statement: Project $\mathbf{b} = [2, 1, 4]^T$ onto the column space of $A = \begin{bmatrix} 1 & 1 \\ 0 & 2 \\ 1 & -1 \end{bmatrix}$.

Step 1 (Check Column Orthogonality):

$\mathbf{a}_1^T\mathbf{a}_2 = (1)(1) + (0)(2) + (1)(-1) = 1 + 0 - 1 = \mathbf{0}$! The columns are mutually orthogonal!

Step 2 (Apply Fast-Track 8 Decoupled Projections):

$$\mathbf{p} = \frac{\mathbf{a}_1^T\mathbf{b}}{\mathbf{a}_1^T\mathbf{a}_1}\mathbf{a}_1 + \frac{\mathbf{a}_2^T\mathbf{b}}{\mathbf{a}_2^T\mathbf{a}_2}\mathbf{a}_2$$

- $\mathbf{a}_1^T\mathbf{a}_1 = 1^2 + 0^2 + 1^2 = 2$; $\mathbf{a}_1^T\mathbf{b} = 1(2) + 0(1) + 1(4) = 6$.
- $\mathbf{a}_2^T\mathbf{a}_2 = 1^2 + 2^2 + (-1)^2 = 6$; $\mathbf{a}_2^T\mathbf{b} = 1(2) + 2(1) - 1(4) = 0$.

Result: $\mathbf{p} = (6/2)\mathbf{a}_1 + (0/6)\mathbf{a}_2 = 3\,[1, 0, 1]^T = \mathbf{[3, 0, 3]^T}$.

Sanity Check on Error: $\mathbf{e} = \mathbf{b} - \mathbf{p} = [2-3, 1-0, 4-3]^T = [-1, 1, 1]^T$.
Check: $\mathbf{a}_1^T\mathbf{e} = -1 + 0 + 1 = 0$ ✓; $\mathbf{a}_2^T\mathbf{e} = -1 + 2 - 1 = 0$ ✓.

Problem 2: Linear Regression via Centered Data Secret Weapon

Chapter 4 & Fast-Track 5

Statement: Fit a line $b = C + Dt$ to four data points: (0, 1), (1, 3), (2, 2), (3, 6).

Fast-Track 5 (Data Centering):

• Compute means: $\bar{t} = (0 + 1 + 2 + 3)/4 = \mathbf{1.5}$; $\bar{b} = (1 + 3 + 2 + 6)/4 = 12/4 = \mathbf{3.0}$.

• Centered predictors $t'_i = t_i - 1.5$: $[-1.5, -0.5, 0.5, 1.5]$.

• Squared sum: $\sum (t'_i)^2 = (-1.5)^2 + (-0.5)^2 + (0.5)^2 + (1.5)^2 = 2.25 + 0.25 + 0.25 + 2.25 = \mathbf{5.0}$.

• Cross-product sum: $\sum t'_i b_i = (-1.5)(1) + (-0.5)(3) + (0.5)(2) + (1.5)(6) = -1.5 - 1.5 + 1.0 + 9.0 = \mathbf{7.0}$.

• **Slope D:** $D = (\sum t'_i b_i) / (\sum (t'_i)^2) = 7.0 / 5.0 = \mathbf{1.4}$.

• **Intercept C:** $C = \bar{b} - D\,\bar{t} = 3.0 - (1.4)(1.5) = 3.0 - 2.1 = \mathbf{0.9}$.

Best-Fit Line: $\mathbf{b = 0.9 + 1.4t}$ (Solved with zero determinants or 2 × 2 matrix inversions!).

Problem 3: Radical-Free Gram-Schmidt & A = QR Decomposition

Chapter 6 & Fast-Track 6

Statement: Compute the QR factorization of $A = \begin{vmatrix} 1 & 2 \\ 2 & 1 \\ 2 & 4 \end{vmatrix}$.

Step 1: First Orthogonal Vector $\mathbf{v_1}$:

$\mathbf{v}_1 = \mathbf{a}_1 = [1, 2, 2]^T$. Norm: $\|\mathbf{v}_1\| = \sqrt{(1 + 4 + 4)} = \sqrt{9} = \mathbf{3}$. So $r_{11} = 3$, $\mathbf{q}_1 = [1/3, 2/3, 2/3]^T$.

Step 2: Projection Coefficient $\mathbf{r_{12}}$:

$r_{12} = \mathbf{q}_1{}^T \mathbf{a}_2 = (1/3)(2) + (2/3)(1) + (2/3)(4) = (2 + 2 + 8)/3 = 12/3 = \mathbf{4}$.

Step 3: Subtract Projection:

$\mathbf{v}_2 = \mathbf{a}_2 - r_{12}\mathbf{q}_1 = [2, 1, 4]^T - 4\,[1/3, 2/3, 2/3]^T = [2 - 4/3, 1 - 8/3, 4 - 8/3]^T = [2/3, -5/3, 4/3]^T$.

Length: $\|\mathbf{v}_2\| = \sqrt{((4 + 25 + 16)/9)} = \sqrt{(45/9)} = \sqrt{5}$. So $r_{22} = \sqrt{5}$, $\mathbf{q}_2 = [2/(3\sqrt{5}), -5/(3\sqrt{5}), 4/(3\sqrt{5})]^T$.

Final Factorization:

$$A = QR = \begin{vmatrix} 1/3 & 2/(3\sqrt{5}) \\ 2/3 & -5/(3\sqrt{5}) \\ 2/3 & 4/(3\sqrt{5}) \end{vmatrix} \begin{vmatrix} 3 & 4 \\ 0 & \sqrt{5} \end{vmatrix}$$

Problem 4: Solving Least Squares via $R\hat{x} = Q^T b$ Back-Substitution

Chapter 7 & Fast-Track 7

Statement: Using the QR factors from Problem 3, find the least squares solution for $\mathbf{b} = [3, 3, 0]^T$.

Step 1: Compute $d = Q^T b$:

• $d_1 = \mathbf{q}_1^T \mathbf{b} = (1/3)(3) + (2/3)(3) + (2/3)(0) = 1 + 2 + 0 = \mathbf{3}$.

• $d_2 = \mathbf{q}_2^T \mathbf{b} = (1/(3\sqrt{5}))\,[2(3) - 5(3) + 4(0)] = (6 - 15)/(3\sqrt{5}) = -9 / (3\sqrt{5}) = \mathbf{-3/\sqrt{5}}$.

Step 2: Fast Back-Substitution on $R\hat{x} = d$:

$$\begin{vmatrix} 3 & 4 \\ 0 & \sqrt{5} \end{vmatrix} \begin{vmatrix} \hat{x}_1 \\ \hat{x}_2 \end{vmatrix} = \begin{vmatrix} 3 \\ -3/\sqrt{5} \end{vmatrix}$$

• Row 2: $\sqrt{5}\,\hat{x}_2 = -3/\sqrt{5} \Rightarrow \mathbf{\hat{x}_2 = -3/5 = -0.6}$.

• Row 1: $3\,\hat{x}_1 + 4(-0.6) = 3 \Rightarrow 3\,\hat{x}_1 = 3 + 2.4 = 5.4 \Rightarrow \mathbf{\hat{x}_1 = 1.8}$.

Optimal Solution: $\hat{x} = [1.8, -0.6]^T$. Completed in 30 seconds without inverting any matrices!

Problem 5: Complete SVD & Rank-1 Decomposition Chapter 9

Statement: Find the SVD of A = $\begin{vmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{vmatrix}$.

Step 1: Form AA^T (2 × 2):

$AA^T = \begin{vmatrix} 17 & 8 \\ 8 & 17 \end{vmatrix} = \begin{vmatrix} 17 & 8 \\ 8 & 17 \end{vmatrix}$.

Step 2: Eigenvalues of AA^T via Fast-Track 9:

$m = (17 + 17)/2 = 17$; $p = 17^2 - 8^2 = 289 - 64 = 225$.
$\lambda = 17 \pm \sqrt{(289 - 225)} = 17 \pm \sqrt{64} = 17 \pm 8 \rightarrow$ **$\lambda_1 = 25$, $\lambda_2 = 9$**.

Singular values: **$\sigma_1 = \sqrt{25} = 5$, $\sigma_2 = \sqrt{9} = 3$**.

Step 3: Eigenvectors U:

For $\lambda_1 = 25$: $(17-25)u_1 + 8u_2 = 0 \rightarrow -8u_1 + 8u_2 = 0 \rightarrow \mathbf{u}_1 = [1/\sqrt{2}, 1/\sqrt{2}]^T$.

By orthogonality: $\mathbf{u}_2 = [1/\sqrt{2}, -1/\sqrt{2}]^T$.

Step 4: Right Singular Vectors V via $\mathbf{v}_i = A^T\mathbf{u}_i / \sigma_i$:

- $\mathbf{v}_1 = (1/5)\, A^T \mathbf{u}_1 = (1/(5\sqrt{2}))\, [5, 5, 0]^T = [1/\sqrt{2}, 1/\sqrt{2}, 0]^T$.
- $\mathbf{v}_2 = (1/3)\, A^T \mathbf{u}_2 = (1/(3\sqrt{2}))\, [1, -1, 4]^T = [1/(3\sqrt{2}), -1/(3\sqrt{2}), 4/(3\sqrt{2})]^T$.
- $\mathbf{v}_3 \in N(A)$: orthogonal to $\mathbf{v}_1$ and $\mathbf{v}_2 \rightarrow \mathbf{v}_3 = [2/3, -2/3, -1/3]^T$.

Result: $A = \sigma_1 \mathbf{u}_1\mathbf{v}_1^T + \sigma_2 \mathbf{u}_2\mathbf{v}_2^T = 5\, \mathbf{u}_1\mathbf{v}_1^T + 3\, \mathbf{u}_2\mathbf{v}_2^T$.

Appendix A: Hiranabe's Five Factorizations & The Matrix World Hierarchy

Factorization	Formula	Matrix Properties	Geometric Meaning
1. Column-Row	$A = CR$	C has r independent cols; R has r independent rows	Basis for C(A) and $C(A^T)$
2. Elimination	$A = LU$	L is lower triangular; U is upper triangular	Gaussian elimination steps
3. Orthogonalization	$A = QR$	Q has orthonormal cols; R is upper triangular	Gram-Schmidt; least squares solver
4. Spectral Theorem	$S = Q \Lambda Q^T$	S is symmetric; Q is orthogonal; Λ is real	Sum of rank-1 orthogonal projection matrices
5. SVD	$A = U \Sigma V^T$	U and V are orthogonal; Σ holds singular values	Two orthonormal bases diagonalizing any linear map

Appendix B: Master Hand-Calculation Cheat Sheet & Formula Summary

Target Operation	Standard Formula	⚡ Fast Paper Shortcut
2 × 2 Matrix Inverse	Gauss-Jordan [M | I]	**Swap** main diagonal, **Negate** off-diagonal, **Divide** by Δ = ad − bc
Symmetric Matrix A^TA	Multiply all n^2 dot products	Compute only diagonal & upper triangle, then **Reflect** across diagonal (saves 50%)
Straight-Line Regression	Solve full 2 × 2 Normal Equations	**Center predictor** $t' = t - \bar{t} \rightarrow A^TA$ diagonalizes → $D = \sum t'b / \sum (t')^2$, $C = \bar{b} - D\,\bar{t}$
Subspace Projection	$P = A(A^TA)^{-1}A^T$	If columns are orthogonal: **$p = \sum (a_i^Tb / a_i^Ta_i)\, a_i$** (No matrix inversion!)
Gram-Schmidt Process	Normalize at every step	**Integer Method:** Keep v_k un-normalized, project with integer ratios, normalize ONLY at the end
Least Squares via QR	$A^TA\,\hat{x} = A^Tb$	**$R\,\hat{x} = Q^Tb$:** Solve strictly via **back-substitution** without inverting any matrices
2 × 2 Eigenvalues	Solve $\lambda^2 - \text{tr}(S)\lambda + \det(S) = 0$	**$\lambda = m \pm \sqrt{(m^2 - \det)}$** where m = tr(S)/2
2 × 2 Eigenvectors	Row reduce (S − λI)	For [(a−λ), b], immediate orthogonal vector is **$[-b, (a-\lambda)]^T$**